

Math 241, F12
Quiz 3

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (6 points) Find parametric equations and symmetric equations for the line.

(a) The line through the origin and the point $(1, 2, 3)$.

$$P_0 = (0, 0, 0)$$

$$V = \langle 1, 2, 3 \rangle - \langle 0, 0, 0 \rangle = \langle 1, 2, 3 \rangle$$

$$\Rightarrow r = \langle 0, 0, 0 \rangle + t \langle 1, 2, 3 \rangle = \langle t, 2t, 3t \rangle$$

$$\text{Parametric equations: } x=t, y=2t, z=3t$$

$$\text{symmetric equations: } \frac{x}{1} = \frac{y}{2} = \frac{z}{3}$$

(b) The line through the point $(1, 0, 2)$ and perpendicular to the plane $3x - 4y + 7z = 12$.

$$P_0 = (1, 0, 2)$$

$$V = \langle 3, -4, 7 \rangle$$

$$\Rightarrow r = \langle 1, 0, 2 \rangle + t \langle 3, -4, 7 \rangle = \langle 1+3t, -4t, 2+7t \rangle$$

$$\text{Parametric equations: } x=1+3t, y=-4t, z=2+7t$$

$$\text{symmetric equations: } \frac{x-1}{3} = \frac{y}{-4} = \frac{z-2}{7}$$

Problem 2. (4 points) Find an equation of the plane through the points $(0, 1, 2)$, $(2, 0, 1)$, and $(1, 2, 0)$.

$$\text{let } P=(0, 1, 2), Q=(2, 0, 1), R=(1, 2, 0)$$

$$\vec{PQ} = \langle 2, -1, -1 \rangle, \vec{PR} = \langle 1, 1, -2 \rangle$$

$$V = \vec{PQ} \times \vec{PR} = \begin{vmatrix} i & j & k \\ 2 & -1 & -1 \\ 1 & 1 & -2 \end{vmatrix} = 3i + 3j + 3k$$

Thus the plane through the three points is given by

$$3(x-0) + 3(y-1) + 3(z-2) = 0$$

$$3x + 3y + 3z = 9$$

$$\underline{x + y + z = 3.}$$